

Domain & Range

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Domain: all the _____ values (x) a function is allowed to use.

Range: all the _____ values (y) the function actually produces.

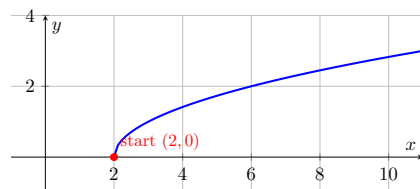
We do together. Find the domain and range of $f(x) = \sqrt{x-2}$.

Step 1. A square root can't hold a negative: set $x - 2 \geq 0$, so the domain is _____.

Step 2. The smallest output is at $x = 2$, where $f(2) = \underline{\hspace{2cm}}$, then it grows.

Step 3. So the range is _____.

See it:



$y = \sqrt{x-2}$ — domain $x \geq 2$, range $y \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Give the domain of $f(x) = \frac{1}{x-5}$.

b) Give the range of $f(x) = x^2 + 1$.

Summary (fill in): Watch two traps: a _____ can't be 0, and a _____ can't hold a negative.

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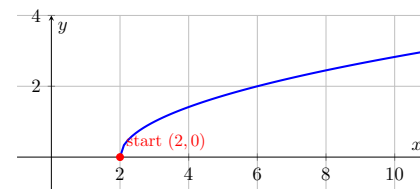
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